

AI-POWERED DIGITAL SAFETY COPILOT

Product Requirements & Architecture Discussion Document

From individual scam detection to collaborative digital safety

Working product thesis

A neutral digital safety layer that analyzes suspicious messages, screenshots, links, QR codes, documents, audio and video, explains the risk in simple language, and brings in a trusted person when the situation is high-risk.

Document status	Team discussion draft
Primary objective	Agree scope, MVP, architecture, safety controls and demo flow
Product name	AI-Powered Digital Safety Copilot / Safety Circle
Operating principle	AI detects and explains; humans verify high-risk situations

1. Executive Summary

The proposed product is an AI-powered Digital Safety Copilot focused on the moment when a user is about to make a potentially unsafe digital decision. The product accepts multimodal inputs such as messages, screenshots, URLs, QR codes, documents, voice and video. It combines deterministic security signals, reputation/context signals and AI reasoning to produce an understandable risk assessment and a recommended safe action.

The key differentiator is the collaborative Safety Circle. Many vulnerable users may recognize fewer modern scam patterns than their children, relatives or trusted contacts. For configurable high-risk scenarios, the Copilot can place the interaction into a safety-review state, warn the user and notify one or more trusted contacts through supported channels. The trusted contact can review a concise evidence summary and recommend a safe action.

Core positioning

Do not position the product as another 'AI scam detector'. Position it as a 'digital decision-safety network' that combines AI detection, explainability, safe verification and trusted-human intervention.

2. Problem Statement

- Scams increasingly combine multiple channels: SMS, WhatsApp, email, websites, QR codes, payment requests, voice calls and video calls.
- Attackers frequently exploit urgency, authority, impersonation, fear, secrecy and emotional pressure rather than relying only on technical vulnerabilities.
- A warning shown only to the recipient can be insufficient when the recipient is inexperienced, stressed or under strong social pressure.
- Modern scams can impersonate a trusted family member, employer, bank, government official, recruiter or service provider.
- Deepfake audio/video introduces a new problem: visual or voice familiarity can no longer be treated as proof of identity.
- Users need a safe verification path, not just a classification such as 'scam' or 'not scam'.
- Trusted contacts can provide a second pair of eyes, but existing security tools often do not make this collaborative workflow the primary product experience.

3. Product Vision

The product should make digital safety understandable and accessible to ordinary users, especially people with lower exposure to modern scams, while allowing trusted people to participate in high-risk decisions.

Capability	What the product should do
Detect	Analyze multimodal content and identify suspicious or high-risk patterns.
Explain	Translate technical findings into plain-language reasons a user can understand.
Verify	Provide a safe way to validate the identity or claim using an independently trusted channel.
Protect	Warn, hold/mark under review where permitted, and recommend a safe next step.
Collaborate	Notify a configured Safety Circle for selected risk levels and situations.
Recover	Support the user when they have already clicked, shared information or paid.

Learn	Turn resolved incidents into anonymized scam-pattern and family-learning signals.
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4. Target Users & Roles

Role	Needs	Key controls
Primary user	Simple answer, understandable explanation, minimal friction	What to analyze; which contacts can be alerted; sharing consent
Trusted contact	Concise context, evidence, ability to advise quickly	Receive alerts; review; mark safe/keep blocked; call user
Safety Circle owner	Configure family policy without taking over the user's device	Risk thresholds; channels; protected scenarios
Enterprise user	Protection from impersonation/payment/social-engineering scams	Organization policy, escalation path, auditability
System administrator	Reliability, security, privacy and model governance	Policies, thresholds, logging, access control

5. Use Case Catalogue

Priority is based on user harm potential, differentiation and demo value.

ID	Use Case	Description	Priority
UC-01	Suspicious SMS/WhatsApp	Courier, KYC, bank, electricity, tax, refund or delivery-payment messages.	P0
UC-02	Fake job/recruiter	Offer letters, registration fees, training fees, fake recruiter links.	P1
UC-03	UPI / QR payment request	Unknown collect request, QR code, refund/payment redirection.	P0
UC-04	Investment / loan scam	Guaranteed returns, urgent deposits, advance fees, fake account managers.	P1
UC-05	Fake authority impersonation	Police, government, court, tax or bank official demanding immediate action.	P0
UC-06	Family/friend impersonation	Known-person impersonation from new number with money request.	P0
UC-07	Deepfake emergency video call	Caller appears to be child/relative and requests urgent money.	P0
UC-08	Voice-clone/voice-note scam	Known-person voice asking for money or sensitive information.	P1
UC-09	Fake customer support	Caller/video caller asks for remote-access app or verification payment.	P1
UC-10	Marketplace scam	Fake buyer/seller, payment screenshot, courier verification link.	P1
UC-11	Fraudulent document	Fake invoice, offer letter, notice, bank/government document.	P1
UC-12	Post-incident recovery	User already clicked, shared information or paid; provide containment guidance.	P0
UC-13	User impersonation	Someone appears to be the user and sends scam requests to their contacts.	P2

UC-14	Family learning	Resolved scam pattern is summarized for the Safety Circle to improve future awareness.	P2
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6. Flagship Scenario: Deepfake Family-Emergency Video Call

Scenario: an elderly user receives a video call from an unknown number. The caller appears and sounds like a close family member and says they are in trouble, need money immediately and do not want the user to tell anyone.

Stage	System behaviour	Expected user experience
1. Trigger	User chooses 'Check this call' or shares a call recording/screenshot where supported.	No cybersecurity expertise required.
2. Media analysis	Analyze available video/audio signals such as face consistency, voice characteristics and audio/video synchronization anomalies.	System reports evidence, not false certainty.
3. Context analysis	Review caller number, time of call, request type, urgency, secrecy language and unusual behaviour.	Context can increase risk even when media looks authentic.
4. Identity verification	Where explicitly configured and technically permitted, attempt verification via an independently trusted contact/channel.	Never rely on a number/link supplied by the suspicious caller.
5. Risk decision	Combine evidence into a confidence-aware risk level.	Example: 'High risk: identity could not be independently verified and multiple social-engineering signals were detected.'
6. Safety Circle	Notify trusted contact for configured high-risk scenarios.	Trusted contact receives concise evidence + recommended action.
7. User protection	Mark/hold the interaction where platform capability permits, or strongly warn the user otherwise.	'Do not send money. Verify through a known contact method.'

Critical requirement

The system must not claim that a deepfake detector proves identity. Deepfake/media signals are evidence. The safer product decision is based on combined media + context + behaviour + trusted-channel verification.

7. Collaborative Safety: Safety Circle

A Safety Circle is a user-controlled group of trusted people who may be notified when configured risk conditions are met.

Feature	Requirement
Contact setup	Primary user can add, remove and verify trusted contacts.
Consent	User explicitly chooses what can trigger an alert and what information is shared.
Risk-based intervention	Low-risk events should not create family notification noise. High/critical events may trigger alerts.
Evidence summary	Alert includes risk, scam category, key evidence and recommended action.
Review actions	Trusted contact can mark 'Safe', 'Keep under review/blocked' or 'Contact user'.

User autonomy	Trusted contact should advise/review; they should not silently take over the user's account.
Channel abstraction	In-app notification is preferred. SMS/messaging integrations are delivery channels subject to platform/API capability and consent.
Escalation	If the trusted contact does not respond, system should not imply approval. It should default to the safe recommendation.
Audit trail	Record alert, review decision and timestamp without unnecessarily retaining full message/video content.
Learning	Resolved incidents can become privacy-preserving pattern signals for future detection.

8. Risk & Intervention Model

Risk	Typical signals	User action	Safety Circle action
LOW	No meaningful red flags; expected sender/context	Normal experience	No alert
MEDIUM	Suspicious language, unfamiliar link, unusual request	Warn + recommend verification	Optional/configurable notification
HIGH	Impersonation + urgency + payment/data request + suspicious destination	Strong warning; pause if supported	Notify trusted contact; review recommended
CRITICAL	High-value payment, deepfake/family emergency, authority intimidation, remote access or secrecy/isolation request	Strong intervention; hold where supported	Immediate alert; trusted verification recommended

9. Safe Action Engine

Every detection should end with an action. A risk score without an action is insufficient.

- Verify through the official app/site or a known contact method.
- Do not use the link, number or payment details supplied by the suspicious message.
- Wait for Safety Circle review for configured high-risk cases.
- Block / report the sender or message where supported.
- If the user already acted, switch to the recovery workflow.
- Provide a 'Why is this risky?' explanation for education and trust.

UX principle

The interface should answer three questions in order: What happened? Why is it risky? What should I do right now?

10. SDLC -> ADLC Transformation for This Product

The product should use the same traditional engineering discipline while adding agentic planning, continuous evaluation, model/prompt governance and autonomous engineering workflows.

Particulars	SDLC (Software Development Lifecycle)	ADLC (Agentic Development Lifecycle)
Driver	Every phase is manually executed by humans.	Agents automate or assist execution of repeatable engineering tasks.
Planning	Scope, budget, requirements and estimates are locked up front.	Goals and product requirements are treated as evolving inputs; agents create/maintain plans and identify gaps.

Requirements	Humans gather and document requirements.	Agents synthesize user stories, acceptance criteria, policy constraints and test scenarios from approved inputs.
Development speed	Phases typically begin after preceding work is reviewed/signed off.	Multiple agent workstreams can run in parallel on isolated tasks.
Design	Architecture and API design are created and reviewed by engineers.	Agents can propose architectures, sequence flows, threat models and test matrices; humans approve architecture decisions.
Coding	Developers implement manually.	Coding agents generate/refactor/test code under guardrails and review requirements.
Testing	QA is commonly a dedicated phase after implementation.	Agents continuously generate tests, run suites, mutate inputs and identify regressions during development.
Evaluation	KPI/performance reviews are often periodic.	Continuous evaluation covers model quality, hallucination, privacy, safety, latency and business outcomes.
Deployment	Release follows approval and release readiness gates.	Agents prepare deployment artifacts, checks and observability dashboards; humans retain release authority for high-impact changes.
Maintenance	Bug fixes and patches are reactive.	Agents observe failures, propose fixes, regenerate tests and maintain runbooks within controlled approval workflows.
Feedback loop	Learning often happens retrospectively.	Telemetry, user feedback and incident outcomes feed continuous improvement and evaluation.

11. Functional Requirements

ID	Area	Requirement	Priority
FR-01	Multimodal intake	System shall accept text, screenshots/images, URLs, QR codes and documents for analysis.	P0
FR-02	Voice/video intake	System should accept permitted audio/video recordings or extracted evidence for impersonation analysis.	P1
FR-03	Content extraction	System shall extract text, URLs, QR payloads, sender identity cues and document metadata where available.	P0
FR-04	Threat/context signals	System shall evaluate reputation, redirect/domain signals, known scam patterns and contextual indicators.	P0
FR-05	Risk engine	System shall produce a risk category, confidence and evidence list.	P0
FR-06	Explanation	System shall explain findings in plain language and avoid unexplained model outputs.	P0
FR-07	Safe action	System shall provide an actionable recommendation tailored to the scenario.	P0
FR-08	Safety Circle	System shall allow user-configured trusted contacts and notification policies.	P0
FR-09	Alert payload	System shall send concise alerts containing risk, evidence, recommended action and review options.	P0

FR-10	Review workflow	Trusted contacts shall be able to review and advise on a flagged event.	P0
FR-11	Deepfake scenario	System should combine media evidence with behavioral/context signals and trusted verification.	P0
FR-12	Post-incident	System shall support a recovery workflow when user indicates they already acted.	P0
FR-13	Feedback	User/trusted-contact feedback shall be captured as a structured signal for evaluation.	P1
FR-14	Auditability	System shall record policy decisions and review events in a privacy-conscious audit log.	P0

12. Non-Functional Requirements

ID	Area	Requirement
NFR-01	Privacy	Minimize retention; redact sensitive information when feasible; make sharing explicit.
NFR-02	Security	Encrypt data in transit/at rest; enforce strong authentication and least privilege.
NFR-03	Trustworthiness	No fabricated evidence. The system must distinguish observed facts, inferred signals and unknowns.
NFR-04	Safety	Never request or expose passwords, OTPs, PINs, private keys or other secrets.
NFR-05	Explainability	High-risk verdicts must provide understandable evidence.
NFR-06	Availability	Core safety check should remain usable during peak alert periods.
NFR-07	Latency	Target a fast response for text/image checks; communicate progress for heavier audio/video analysis.
NFR-08	Localization	Design content model to support English plus Indian languages in future.
NFR-09	Observability	Capture model, rules, latency, confidence and workflow telemetry without unnecessarily retaining content.
NFR-10	Governance	Maintain versioning for prompts, models, rules and threat-intelligence sources.

13. Logical Architecture

Layer	Components	Responsibility
Experience	Web/mobile UI; share/upload; Safety Circle dashboard	Collect evidence, show result, capture consent and review decisions.
API / Orchestration	API gateway; workflow/orchestration service	Route requests, enforce policy and coordinate analyzers.
Extraction	OCR; QR decoder; URL normalization; document parser; audio/video feature extraction	Turn raw submissions into analyzable signals.

Security signals	URL/domain reputation; redirect analysis; scam-pattern rules; sender/context signals	Provide deterministic and external evidence.
AI reasoning	Multimodal LLM / classifiers	Summarize evidence, identify social-engineering patterns and generate explanation.
Risk engine	Weighted scoring; policy rules; confidence calibration	Combine signals and determine risk + intervention.
Safety Circle	Policy engine; notification service; contact verification; review UI	Trigger alerts and collect trusted-contact decisions.
Recovery	Incident workflow; safe-channel guidance; evidence preservation	Help user after potential exposure/payment.
Evaluation	Offline benchmark; red-team tests; production telemetry	Continuously measure quality, safety and drift.
Governance	Model/prompt/rule registry; audit logs; access control	Control changes and support traceability.

14. Core Data Objects

Object	Key fields	Notes
SafetyUser	user_id, locale, consent_policy, risk_preferences	Primary product user.
TrustedContact	contact_id, relationship, verified_channel, notification_preference	Must be explicitly verified/consented.
SafetyPolicy	risk_threshold, protected_scenarios, action_mode, escalation_policy	Defines when collaboration activates.
SafetyEvent	event_id, type, timestamp, source_type, risk, evidence_summary	Core unit of analysis.
Evidence	evidence_id, source, signal_type, confidence, observed_value	Store the minimum required evidence.
Review	event_id, reviewer_id, decision, timestamp, note	Captures collaborative action.
ModelDecision	model_version, rule_version, prompt_version, score, uncertainty	Supports evaluation and audit.
Incident	event_id, incident_state, recommended_actions, outcome	For post-incident recovery.

15. Agentic Development & Operations Workflow

Agents should augment engineering rather than become an unchecked production authority.

ADLC Activity	Example agent responsibility	Human gate
Requirements agent	Generate user stories, acceptance criteria, edge cases and traceability matrix.	Product owner approves scope and safety requirements.
Architecture agent	Propose service boundaries, data flows and threat model.	Lead engineer approves architecture.
Coding agent	Implement isolated stories, tests and refactors.	Developer reviews code and security findings.
Test agent	Generate adversarial samples, regression tests and multimodal test cases.	QA/engineer approves critical suites.
Evaluation agent	Run benchmark sets, compare model versions, detect drift and unsafe outputs.	Model owner approves promotion.
Observability agent	Summarize incidents, latency changes and failure clusters.	On-call engineer approves remediation.
Documentation agent	Keep requirements, API docs, runbooks and decision logs synchronized.	Owner signs off release documentation.

16. Hackathon MVP Scope

Recommended MVP

Build one complete journey exceptionally well: suspicious message / screenshot -> risk assessment -> evidence -> Safety Circle alert -> trusted contact decision -> safe action. Add the deepfake family-emergency flow as the second flagship journey if time permits.

MVP feature	In scope	Demo priority
Screenshot/message analysis	OCR + scam classification + evidence + explanation	P0
URL analysis	Normalize URL + basic reputation/context checks	P0
QR analysis	Decode QR and analyze destination	P0
Safety Circle	Add trusted contact + high-risk notification	P0
Review workflow	Trusted contact sees alert + Keep Safe/Keep Blocked/Call User	P0
Deepfake video scenario	Recorded clip or simulated analysis with contextual signals	P1
Post-incident mode	Already clicked/paid -> guided recovery	P1
Multilingual output	One additional Indian language	P2
Real-time WhatsApp/SMS interception	Actual interception depends on OS/platform APIs; use simulated/demo integration if needed	P2

17. Acceptance Criteria for the Demo

- The system produces a risk level and at least three understandable reasons for each high-risk demo case.
- For a configured high-risk event, the trusted contact receives an alert with sufficient context to act without opening the raw message.
- The user sees a clear recommended next action and a safe verification method.
- The system demonstrates a legitimate message that is not incorrectly blocked by default.
- The deepfake scenario does not claim certainty; it explicitly communicates uncertainty and recommends independent verification.
- The Safety Circle cannot be added silently; contact verification and consent are required.
- No password, OTP, PIN or secret is requested as part of analysis.
- The product can explain what information was shared with the trusted contact.
- Core demo requests complete within an acceptable time for the chosen modality, with progress feedback for heavier analysis.

18. Success Metrics

Metric	Definition	Why it matters
Detection precision/recall	Quality by scam type	Avoid false confidence.
Safe-action rate	% of users selecting recommended safe action after analysis	Measures real user outcome.
Intervention yield	% of high-risk events where Safety Circle review changes outcome	Measures collaborative value.
False-positive rate	Legitimate cases incorrectly escalated/blocked	Protects user trust.
Explanation satisfaction	Human rating of clarity/relevance	AI explanation is part of product.
Verification success	% of cases independently verified through trusted channel	Core safety mechanism.
Time to guidance	Submission to actionable recommendation	Users may be under pressure.

Privacy leakage	Amount of sensitive content retained/shared	Critical trust metric.
Model drift	Performance change over time / new scam patterns	Supports ADLC evaluation.

19. Threat Model & Abuse Cases

- Attacker submits a legitimate-looking artifact to manipulate the model into a false 'safe' verdict.
- Attacker attempts prompt injection through a document, image or message.
- Attacker impersonates a trusted contact or abuses the notification channel.
- A malicious trusted contact misuses the Safety Circle to control or harass the user.
- False positives cause users to ignore future warnings.
- Users expose secrets by uploading credentials, OTPs or screenshots containing sensitive data.
- Model bias or insufficient local scam examples reduces detection quality for regional languages and patterns.
- Agent-generated code or configuration introduces a security regression.

Mitigation should combine deterministic policy enforcement, adversarial evaluation, human approval gates, rate limits, data minimization and explicit auditability.

20. Open Questions for Team Discussion

- Which channels will the hackathon prototype actually support: web upload, Android share sheet, simulated SMS/WhatsApp, browser extension, or a combination?
- Should 'under review' mean a true blocking/hold state or only a user-visible warning in the MVP?
- Which trusted channels are technically permitted for independent identity verification?
- How do we verify a trusted contact without creating a new social-engineering attack surface?
- What minimum amount of content should be shared with the trusted contact?
- Should the user be allowed to override a trusted-contact recommendation on HIGH risk events?
- Which India-specific scam scenarios are our benchmark set?
- How will we handle multilingual OCR and mixed-language messages?
- Which deepfake signals can we demonstrate credibly within hackathon time?
- What is our fallback when threat-intelligence APIs are unavailable?
- What data do we retain for evaluation, and for how long?
- Which actions are informational only versus actions requiring a user confirmation or external service integration?

21. Team Decision Log

Decision	Options considered	Chosen option	Owner	Date	Notes
MVP channels					
Risk thresholds					
Safety Circle notification channel					
Deepfake demo scope					
Threat-intelligence source					
LLM/model choice					
Data retention					

22. Recommended Product Narrative

Problem

Scammers increasingly use trusted identities, realistic media and emotional pressure. A warning shown only to the recipient may arrive too late.

Solution

An AI Copilot that analyzes the evidence, explains the risk and tells the user exactly how to verify safely.

Differentiator

Safety Circle: when risk is high, the Copilot can bring in a trusted human before a risky decision becomes irreversible.

Flagship experience

A deepfake family-emergency video call where the Copilot does not try to 'prove' the deepfake; it combines media, context and trusted-channel verification to stop a pressured payment.

North-star outcome

Prevent unsafe digital decisions, while preserving user autonomy, privacy and trust.

Appendix A. SDLC / ADLC Phase Mapping

Phase	SDLC activity for this product	ADLC enhancement	Primary artifact
Planning	Define target users, hackathon scope, constraints and success metrics.	Agent-assisted opportunity analysis and backlog generation.	Product charter
Requirements	Document functional/NFRs and acceptance criteria.	Agents generate edge cases and traceability.	PRD/requirements
System Design	Architecture, threat model, data model and integration choices.	Agents propose alternatives and test architecture assumptions.	Architecture decision record
Implementation	Build UI, services, risk engine, notifications and integrations.	Coding/test agents work on isolated stories.	Source + tests
Testing & QA	Unit, integration, security and UX testing.	Continuous adversarial/model evaluation.	Evaluation report
Deployment	Package and deploy demo services.	Agents validate configuration/observability; human approves release.	Release checklist
Maintenance	Bug fixes, model/rule updates and support.	Agents cluster incidents, propose fixes and regenerate regression tests.	Runbooks/change records
Evaluation	Periodic KPI/performance review.	Continuous quality, safety, drift and feedback evaluation.	Evaluation dashboard

Appendix B. Reference User Journey

Step	Primary user	System	Safety Circle
1	Receives suspicious content	Detects / analyzes when user asks for a check	—
2	Asks 'Is this safe?'	Extracts evidence + scores risk	—
3	Sees HIGH risk	Explains why + safe action	—
4	Waits / chooses review	Applies policy	Trusted contact alerted
5	Trusted contact reviews	Provides evidence and review options	Marks safe / keep blocked / contact user

6	User acts safely	Records resolution	Review event stored
7	Future case	Uses feedback to improve evaluation	Family learning can optionally surface pattern